

RESEARCH STATEMENT

by LILI JU

I have worked on a wide range of projects involving mathematical modeling, algorithm development, numerical analysis, and computer implementation for various problems arising from science and engineering. My research efforts have encompassed areas of classic numerical partial differential equations, deep learning methods, and high-performance computing with applications to materials science, geological science, computer vision, biological science, and data science. Since 2006 my work has been continuously supported by many research grants (a total amount of around 2.81 million dollars) from U.S. National Science Foundation¹ and U.S. Department of Energy². I have published over 140 papers in many first-tier academic journals³ and top conference proceedings⁴ in computational and applied mathematics and interdisciplinary sciences, with a Google Scholar record of about 5000 citations and H-index 41. I have collaborated with researchers and scientists not only from peer universities and institutions but also from national laboratories such as Los Alamos National Laboratory and Sandia National Laboratories. A summary of my research work (completed and current) is presented below.

1 Exponential integrators for stiff evolution equations

- *Developed an abstract analysis framework for studying maximum bound principles (MBP) of many well-known semilinear parabolic and phase field equations and their numerical discretizations, and proposed unconditionally MBP-preserving exponential integrator based schemes with high-order temporal accuracy. See [50, 61, 62, 65, 63, 72].*
- *Developed and implemented accurate, stable and efficient numerical schemes for many phase field models, including domain decomposition-based localized exponential time differencing method (LETD). Our LETD-based extreme-scale parallel phase field simulations of 3D coarsening dynamics on the “Sunway TaihuLight” supercomputer won a finalist of 2016 ACM Gordon Bell Prize in high-performance computing. See [28, 31, 34, 35, 41, 42, 50, 37, 46, 58, 60].*

Many important physical phenomena are modeled by semilinear or nonlinear stiff evolution equations. Some typical examples include Allen-Cahn and Cahn-Hilliard equations for phase transition, the advection-diffusion equation, Burger’s equation and Navier-Stokes equation for fluid dynamics, the phase field elastic bending energy model for vesicle membranes, the epitaxy growth model for thin film deposition, the phase field crystal equation for atomic crystals formulation, the Ginzburg-Landau model for superconductivity and so on.

Fast compact exponential time differencing – In [28, 35] we proposed an explicit exponential time differencing (ETD) multistep method and its fast implementation for the solution of Allen-Cahn equations. The method combines decompositions of compact spatial difference operators on a regular mesh with stable and accurate exponential time integrators and efficient discrete FFT-based algorithms. It can deal with stiff nonlinearity and both homogeneous and inhomogeneous

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³SIAM Review, SIAM Journal on Numerical Analysis, Mathematics of Computation, Numerische Mathematik, SIAM Journal on Scientific Computing, Journal of Computational Physics, Computer Methods in Applied Mechanics and Engineering, Journal of Scientific Computing, Communications in Computational Physics, IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Image Processing, Journal of Geophysical Research, The Cryosphere, and etc

⁴AAAI, CVPR, ICCV, ECCV, MSML, and etc

boundary conditions of different types based on multistep approximations and analytic evaluations of time integrals. We extended in [31] this method for stable and accurate long-time dynamics simulations of the Cahn-Hilliard equation in rectangular domains, in which two linear splittings were also incorporated into the schemes to improve numerical and energy stability in time. To efficiently simulate the Allen-Cahn dynamics of the constrained phase field elastic bending energy model, we developed in [34] a modified augmented Lagrange multiplier (ALM) approach and combined it with the stabilized ETD Runge-Kutta schemes for time integration and the spectral discretizations in space. We studied in [40] the ETD multistep method for solving the gauge system of incompressible viscous flows governed by the incompressible Stokes or Navier-Stokes equations. A compact representation of the algorithm for problems on rectangular domains was developed, which makes FFT-based solvers again available for the resulting fully discretized system. We also studied the ETD methods for the compressible Navier-Stokes equation in [43], where a new second-order exponential time integrator scheme, named the “Predictor-Corrector Exponential” scheme was developed and applied to fluid dynamics problems, which provides accurate fast time marching with large time step sizes. In [41], we proposed a class of ETD schemes with spectral discretization in space for solving the epitaxial growth model without slope selection.

Localized exponential integrators – Direct parallelization of global exponential integrators on the whole domain is often very hard to be scalable on massively distributed systems due to intensive data communications needed by FFT in the compact representation or by the Krylov subspace algorithms for products of matrix exponentials and vectors. We developed and implemented in [37] a highly scalable compact localized ETD method based on domain decomposition for extreme-scale parallel phase field simulations of 3D coarsening dynamics on the “*Sunway TaihuLight*” supercomputer. The main idea is to use domain decomposition techniques to reduce the size of the problem, so that one instead only solves a group of smaller-sized subdomain problems simultaneously using the locally computed products of matrix exponentials and vectors. This work won us one of the finalists in the supercomputing’s most prestigious competition, 2016 ACM Gordon Bell Prize in high-performance computing⁵. With the diffusion equation as the model problem, we later developed and analyzed in [46, 58, 60] respectively some overlapping and nonoverlapping localized ETD schemes and solution algorithms for diffusion equations and semilinear parabolic equations, which are to serve as the first step toward building a solid mathematical foundation for localized exponential integrators.

Maximum bound principle-preserving exponential integrators – In [42, 50] and later in the *SIAM review* article [61], we considered a practically desirable property for a class of semilinear parabolic equations of the abstract form $u_t = Lu + f[u]$ with L being a linear dissipative operator and f being a nonlinear operator in space, namely a maximum bound principle (MBP), in the sense that the time-dependent solution u preserves for all time a uniform pointwise bound in absolute value imposed by its initial and boundary conditions. We studied an analytical framework for some sufficient conditions on L and f that lead to such a MBP for the time-continuous dynamic system of infinite or finite dimensions. Then, we utilized a suitable ETD approach with a properly chosen generator of contraction semigroup to develop first- and second-order accurate temporal discretization schemes, that satisfy the maximum bound principle unconditionally in the time-discrete setting. Extensions to vector- and matrix-valued systems were also discussed. We demonstrated that the abstract framework and analysis techniques developed here offer an effective and unified approach to study the maximum bound principle of the abstract evolution equation, that covers a wide variety of well-known models (such as the Allen-Cahn type gradient flows) and their numerical discretization schemes. Later, we further developed in [62, 63] some high-order (up to fourth-order) integrating factor Runge-Kutta (IFRK) schemes which can preserve the MBP conditionally or unconditionally. More recently, we proposed and analyzed [72] a novel generalized SAV-exponential integrator schemes for the Allen-

⁵<http://sc16.supercomputing.org/2016/08/23/finalists-compete-prestigious-acm-gordon-bell-prize-high-performance-computing/index.html>

Cahn flows and in [65, 74, 79] some ETD schemes for two types of mass-conserving Allen-Cahn equations.

2 Ocean and ice sheet modeling and simulations

- *Developed and implemented a prototype 3D thermo-mechanical coupled finite element full-Stokes ice-sheet model to explore ice-sheet changes under realistic conditions. Our model is the core component for the Stoke model in FELIX (Finite Element Land Ice eXperiments) supported by U.S. Department of Energy PISCEES project. See [21, 22, 26, 27, 38].*
- *Developed and analyzed conservative explicit local time-stepping (LTS) schemes with high-order accuracy for shallow water equations and hyperbolic conservation laws, which are highly scalable for parallel computing. See [48, 55].*

Numerical modeling of land ice evolution has been a subject of growing interest because of the crucial role land ice plays in global sea level and other parts of the climate system. Ocean modeling for describing physical and thermodynamical processes in oceans has always been the core component of the earth system models.

Parallel FEM ice-sheet modeling – Nonlinear 3D Stokes flow is the gold standard among conceptual models for ice sheet dynamics. The current widely-used shallow-ice, shallow-shelf, L1L2, and higher-order approximations are all obtained as reduced forms of the 3D Stokes model by means of scaling analysis, but in many situations, with an attendant loss of fidelity. In [21, 27, 26], we developed a prototype 3D thermo-mechanical coupled finite element full-Stokes ice-sheet model to explore ice-sheet changes under realistic conditions. The ice-sheet dynamic problem is theoretically an anisotropic p -Stokes problem, and the difficulty of solving it comes from anisotropy and the drastically changing viscosity. We proposed a hybrid Picard-Newton method for solution of the resulting nonlinear system. Parallel algorithms based on domain decomposition were employed to solve the respective sub-models using Message Passing Interface (MPI). We also introduced the block preconditioner to solve the linear systems using the proper Schur complement for variable coefficients and algebraic multigrid solver for velocity subdomain problems. Our computational model was regarded as “*a new generation of full-stress ice-sheet models incorporated the physics needed to reproduce such processes*” by Alley and Joughin in their 2012 *Science* paper. The manufactured solutions are often used for the verification of computational ice-sheet models, and we proposed in [22] the first correct 3D manufactured solutions for full-Stokes ice-sheet problem, which has been accepted by many ice-sheet modeling groups as the basic verification tools for their models.

Multiresolution and multirate ocean simulations – We reported and compared some simulation results [12, 17, 20] based on the shallow water equations which elucidate both the potential benefits of the variable-resolution SCVT meshes and the challenges ahead. It has been found by the oceanographers that the conservation of mass, vorticity and energy are essential in the long-term ocean simulations, and many spatial discretization schemes are proposed to maintain these conservation properties. In [45] we proposed conservative finite volume schemes for solving the quasi-geostrophic equation on coastal-conforming unstructured primal-dual meshes. In [64, 73] we developed energy-and-ensrophy conserving numerical schemes by approximating the Hamiltonian formulation of the inviscid shallow water flows based on the vorticity-divergence variables. In [48] we developed conservative explicit local time-stepping (LTS) schemes with second and third order accuracy for the shallow water equations. The time integration is designed based on the strong stability preserving Runge-Kutta (SSPRK) methods, but different time step sizes can be used in different regions of the spatial domain through the coupling of coarse-fine time discretizations on the interface, and are only restricted by respective local CFL conditions. Moreover, they inherit the natural parallelism of the original explicit global time-stepping schemes. These results could be leveraged by many coastal and

unstructured fluids models to increase the scientific capability of the many research codes that are currently in use. In addition, we extended these LTS schemes to time stepping of general hyperbolic conservation laws with DG approximations in space in [55]. On the other hand, to treat the multiple time scales of ocean dynamics efficiently, the baroclinic-barotropic dynamics splitting technique has been widely employed for solving the primitive equations in ocean modeling. In [69] we developed two high-order multirate explicit time-stepping (SSPRK-SE) schemes for such split system. The proposed schemes allow for a large time step for advancing the 3D baroclinic mode and a small time step for predicting and correcting the 2D barotropic mode, so that each of the two mode solvers only need to satisfy their respective CFL conditions to maintain overall numerical stability while still achieving high-order accuracies. Meanwhile, the sea surface height perturbations is reconciled with carefully calculated flux deficits and velocity adjustments at all stages of each time step to resolve the numerical inconsistency issue due to the mode splitting.

3 Nonlocal models and numerical computation

- *Developed a general framework for a posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamics models, and studied the connection between nonlocal a posteriori error estimation and classical local estimation. See [23, 29].*

There have been much interests in nonlocal models given in the form of partial integral equations in the past decades. Existing studies range from applications in nonlocal mechanics, diffusion, image analysis, biology to mathematical analysis and computational simulations.

Linear peridynamics and nonlocal convection-diffusion modeling – Peridynamics (PD) models were first introduced by Silling in 2000 and has become an attractive emerging tool for the multi-scale material simulations involving crack growth or damage. It is an integral-type nonlocal continuum theory which provides a more general set-up than the classical PDE based elasticity theories for physical problems with discontinuities; on the other hand it can also be viewed as a continuum version of molecular dynamics. In [23], we obtained some results on a posteriori error analysis of finite element methods for solving linear bond-based nonlocal diffusion and PD models. In particular, we aimed to propose a general abstract framework for a posteriori error analysis of the PD problems. A posteriori error estimators were consequently prompted, and the connection between nonlocal a posteriori error estimation and classical local estimation is studied. In [29], we proposed and compared several nonlocal models for convection-diffusion problems and studied their finite element approximations by piecewise constant and piecewise bilinear basis functions. Through analysis and numerical experiments, We demonstrated the advantage of the upwind models which are more consistent with the physical nature and allow more stable and robust numerical approximations. Then in [39], we proposed a nonlocal convection-diffusion model, in which the convection term is constructed in a special upwind manner so that mass conservation and maximum principle are maintained in any space dimension. A quadrature-based finite difference discretization is developed to numerically solve the nonlocal problem and is shown to be asymptotically compatible.

Numerical methods for nonlocal diffusion problems – In [32] we proposed nodal-type quadrature rules for approximating hypersingular integrals and their applications to numerical solution of finite-part integral equations and nonlocal diffusion problems. Later in [33, 44] we proposed effective and accurate quadrature rules for computing these double integrals for one-dimensional nonlocal problems; in particular, for problems with highly singular kernels, the corresponding inner integrals can be first evaluated exactly in our method, and the outer one then will be approximated by some popular quadrature rules. In [47, 49], we designed and analyzed a novel DG method for solving the one-dimensional time-dependent nonlocal diffusion problem. The key idea of the method is the introduction of an auxiliary variable, analogous to the classic local discontinuous Galerkin (LDG) method but with some nonlocal extensions. Theoretical analysis shows that the proposed semi-discrete DG

scheme is L^2 -stable, convergent, and asymptotically compatible. This latter feature implies that the classical DG approximation of the local diffusion problem can be recovered in the zero horizon limit.

4 Deep learning and data science

- *Developed a novel method of dimension reduction via learning level sets for function approximation, which utilizes a pseudo-reversible neural network and a synthesized regression, which outperforms popular conventional and deep learning methods. See [80].*
- *Developed a set of convolutional neural networks for learning depth and stereo information from multimodal images/videos, which significantly promote the state-of-the-art performance of these tasks. See [52, 53, 54, 56, 57].*
- *Developed an effective end-to-end neural network for unsupervised nighttime semantic segmentation, which performs a one-stage adaptation from a labeled daytime dataset to unlabeled day-night image pairs. See [67, 78].*

Deep Learning is a subset of machine learning that uses mathematical functions to map the input to the output. These functions can effectively extract non-redundant information or features from the data, which enables them to form a relationship between the input and the output. Modern deep learning models use artificial neural networks (ANN) for representation of the functions. We are concerned with a new array of integrated deep learning methods that can reliably and efficiently extract information from scientific data and infer physics for accelerating scientific discovery.

Learning PDE solutions – Inspired by the rapidly growing impact of deep learning techniques, many deep learning based methods have developed for numerical solution of PDEs and attracted lot of attention in the past couple of years. In [66] we designed a neural network model in the unsupervised fashion, called “IFNN”, based on a special implicit form for learning solution of scalar hyperbolic conservation laws. Our IFNN does not directly process the target PDEs, instead it deals with the nonlinear equations characterizing implicitly the solution of the PDEs. In [76] we proposed a novel neural network method for learning the Green’s functions of linear reaction-diffusion equations, which can be applied to fast numerical solution of the corresponding PDE problems with various source terms and Dirichlet boundary values without network retraining. In [71], we proposed a deep learning method “GW-PINN” without labeled data for solving groundwater flow equations with wells was proposed. A locally refined sampling strategy is adopted in our GW-PINN to generate the consistent spatial sampling points for problems with strong hydraulic head change, and then combined with an appropriate temporal sampling scheme to obtain the final spatial-temporal sampling points.

The popularity of deep convolutional autoencoders (CAEs) has engendered new and effective reduced-order models (ROMs) for the simulation of large-scale dynamical systems. Despite this, it is still unknown whether deep CAEs provide superior performance over established linear techniques or other network-based methods in all modeling scenarios. In [70], we investigated the effect of autoencoder architecture on its associated ROM by comparing the deep CAEs against two alternatives: a simple fully connected autoencoder and a novel graph convolutional autoencoder. It is shown that the superior autoencoder architecture for a given ROM application is highly dependent on the size of the latent space and the structure of the snapshot data, with the proposed architecture demonstrating benefits on data with irregular connectivity when the latent space is sufficiently large.

Level set learning and function approximation – In [80] we tackle the problem of finding an approximation of a function, given the information of its values and gradients on a set of randomly chosen sample points. Due to the curse of dimensionality and the limitation on training data, approximating high-dimensional functions is a very challenging task even for powerful deep neural networks (DNN). An effective way to alleviate such difficulty is to reduce the input dimension of the target function by identifying some of its intrinsically low-dimensional structures. We proposed a

model “DRILLS” for dimension reduction via learning level sets and function approximation. Our method contains two major components: one is the pseudo-reversible neural network (PRNN) module that effectively transforms high-dimensional input variables to low-dimensional active variables, and the other is the synthesized regression module for approximating function values based on the transformed data in the low-dimensional space. The PRNN not only relaxes the invertibility constraint of the nonlinear transformation by some existing methods, but also adaptively weights the influence of each sample and controls sensitivity of the function to the learned active variables. The synthesized regression uses Euclidean distance in the input space to select neighboring samples, whose projections on the space of active variables are used to perform local least-squares polynomial fitting.

Depth estimation from images/videos – Depth estimation from multimodal images has important applications in diverse areas, including 3D reconstruction, autonomous driving, augmented reality and robot navigation. In recent years, the accuracies of many computer vision tasks have been greatly improved by using deep learning techniques based on convolutional neural networks (CNNs). We developed two neural networks: one is a semantic stereo network in [52], “SSPCV-Net”, which uses pyramid cost volumes of semantic and spatial information on multiple levels to learn depth from stereo pairs of images, and the other is a network with end-to-end adversarial training in [53], “SC-GAN”, which uses spatial correspondence based on Smolyak sparse grids to learn depth from monocular video frames without estimating the camera pose and pose change over time. In [54], we proposed a new end-to-end deep learning network of P-MVSNet for multi-view stereo (MVS) based on isotropic and anisotropic 3D convolutions. Our P-MVSNet consists of two core modules: a patch-wise aggregation module learns to aggregate the pixel-wise correspondence information of extracted features to generate a matching confidence volume, from which a hybrid 3D U-Net then infers a depth probability distribution and predicts the depth maps. Later we further developed the attention-aware MVS (“AttMVS”) in [56] and the mesh-guided MVS (MG-MVS) in [57].

Cross-domain semantic segmentation – The goal of semantic segmentation is to classify each pixel of images based on the pre-defined classes. In the era of deep learning, convolutional neural networks largely improve the accuracy and efficiency of semantic segmentation. However, this success is still achieved with limitations, for example, a large-scale labeled dataset is required for training while the labeling process for this task is quite labor-intensive and tedious and the trained deep networks can get promising results when testing on the same domain but might suffer from a large performance drop when testing on different domains. In [75], we proposed a new one-shot unsupervised domain adaptation (OSUDA) method for semantic segmentation, which does not require additional data to pre-train a style-transfer module and explicitly synthesizes stylized images for semantic segmentation. In [67, 78], we tackled the problem of semantic segmentation for nighttime images that plays an equally important role as that for daytime images in autonomous driving but is also much more challenging due to very poor illuminations and scarce annotated datasets. We proposed a novel end-to-end neural network for unsupervised nighttime semantic segmentation, which performs a one-stage adaptation from a labeled daytime dataset to unlabeled day-night image pairs. In [77] we proposed a domain adaptation method for video semantic segmentation, and argued that it is not necessary to use a labeled video dataset as the source since the temporal continuity of video segmentation in the target domain can be estimated and enforced without reference to videos in the source domain.

5 Centroidal Voronoi tessellations

- *Proposed the concept of constrained centroidal Voronoi tessellation (CCVT) on surfaces/manifolds and developed some effective construction algorithms. CCVTs have been widely used in the computer graphics community and CCVTs on the surface of the sphere (called “SCVT”) have become standard meshes used by MPAS (Model for Prediction Across Scales) supported by U.S. Department of Energy. See [2, 12, 20].*

- *Proved the nondegeneracy of the classic Lloyd algorithm for computing CVTs and showed that this algorithm is globally convergent in one dimension and weakly globally convergent in higher dimensions. See [7, 10].*
- *Developed a family of CVT-based methods for image segmentation, such as Edge-Weighted CVT (EWCVT), Local Variation EWCVT, Multichannel EWCVT, Hierarchical EWCVT and Fuzzy EWCVT. See [14, 19, 24, 30, 36].*

A comprehensive study of *centroidal Voronoi tessellation* (CVT) was first provided by Du, Faber and Gunzburger in their 1999 SIAM Review article. While the CVT concept initially was phrased as a model and method for optimal point distributions and spacial tessellations of regions/volumes or within sets of discrete data, the generality and universality of CVTs have also made it widely applicable in many areas of science and engineering. CVTs possess an optimization property that can be used as a basis for an alternate definition, where the target energy to be minimized is often naturally associated with quantities such as *quantization error*, *variance* and *cost* in practical applications.

Construction algorithms and analysis – We rigorously proved in [7] the global convergence of the well-known Lloyd algorithm for computing CVTs in one dimension for any positive and smooth density function based on compactness and non-degeneracy of the Lloyd iteration, and later further improved this result in [10] by showing the weak global convergence of the Lloyd iteration in higher dimensions provided that the domain Ω is a convex and bounded set and the density is a L^1 function. We proposed in [1] a probabilistic method for determining CVTs that can be viewed as a generalization or combination of the two classic methods, Lloyd algorithm and k -mean algorithm. It is very important to point out that these probabilistic algorithms do not require, at any stage, the construction of Voronoi diagrams nor the determination of the centroids of the Voronoi cells. Moreover, the algorithms are highly amenable to fully scalable parallelization.

CVTs on surfaces/manifolds – In [2] we generalized the CVT concept and algorithm to surfaces/manifolds by giving a precise definition of *constrained centroidal Voronoi tessellation* (CCVT) and derived a number of properties for such tessellations. Since then CCVTs have been popularly used and studied in the computer graphics and meshing community. A CCVT is specially called *spherical centroidal Voronoi tessellation* (SCVT) when the underlying manifold is the surface of a sphere, which has become the standard meshes used by MPAS (Model for Prediction Across Scales) [20], a collaborative project for developing atmosphere, ocean and other earth-system models (<https://mpas-dev.github.io>) supported by Department of Energy.

CVT-based mesh optimization and adaptivity – A major advantage of CVT-based adaptive computations is that we can explicitly determine a density function based on some posteriori error estimator and the CVT mesh sizing property to generate new meshes so that the errors of the approximate solutions will be equally distributed over the elements in an optimal way. We developed some CVT-based adaptive algorithms for solving second-order elliptic PDEs in [9, 15] and the Navier-Stokes equations in [11]. We also proposed an adaptive anisotropic CVT-meshing scheme based on metric tensor information for steady convection-dominated problems in [13].

CVT-based image segmentation – In the image processing context and in its simplest form, the basic CVT method reduces to the well-known k -means clustering technique based on intensity similarity. By viewing the latter within the CVT context, we proposed several CVT algorithms in [8] for image compression and segmentation applications. However, the classic CVT model lacks the regularity term, thus often produces too many details during segmentation and fails to accurately detect the target objects in heavily noisy images. We then developed a model of *edge-weighted centroidal Voronoi tessellation* (EWCVT) in [14] and its multichannel version in [24] for image segmentation. EWCVT beautifully combines the global color/intensity information and the physical information by adding a regularity term in the original CVT model, and can handle very sophisticated situations. We later developed a *local variation and edge-weighted centroidal Voronoi tessellation* (LVEWCVT)

model in [19] that improves EWCVT by replacing the classic clustering energy term by the local variation energy in order to handle images with background inhomogeneity. In addition, we developed a hierarchical EWCVT (HEWCVT) model in [30] for dealing with superpixel/supervoxel segmentation and a fuzzy EWCVT model in [36].

6 Other works

Numerical methods for geometric PDEs – My work in this area mostly focuses on finite volume and finite element methods for problems defined on surfaces/manifolds, the so-called geometric PDEs. In [3] we developed a Voronoi-based finite volume scheme for discretization of convection-diffusion problems on the sphere based on spherical Voronoi tessellations (SVTs). The Voronoi-based finite volume scheme was shown to produce first-order accurate approximations with respect to a mesh-dependent discrete first-derivative norm. We later rigorously prove that the L^2 error of the approximation is of quadratic order when the underlying Voronoi mesh is given by a SCVT [6]. In [5] we proposed a novel finite volume discretization scheme for approximations of the reduced Ginzburg-Landau phenomenological model for superconducting hollow spheres based on SCVT and used to study the behavior of the quantized vortices [4] on a spherical geometry. The scheme utilizes the properties of the SCVT and the discrete gauge invariance automatically is satisfied to accurately compute the various solution branches, vortex nucleation patterns and vortex dynamics on the sphere. In [18], we considered the phase separation on general surfaces by solving the nonlinear Cahn-Hilliard equation using a finite element method. A fully discrete approximation scheme was introduced, and then we established a priori estimates for the discrete solution that does not rely on any knowledge of the exact solution beyond the initial time.

Domain decomposition methods for Helmholtz equation – In [51], we proposed and analyzed an additive domain decomposition method (DDM) for solving the Helmholtz equation with the Sommerfeld radiation condition in $\mathbb{R}^n$. The computational domain is partitioned into structured subdomains along all spatial directions, and each of the subdomains contains an overlapping region for source transferring through the perfectly matched layer (PML) technique. At each step all subdomain PML problems are solved concurrently, then the horizontal, vertical and corner directional residuals on each subdomain are passed to its corresponding neighbor subdomains as the sources for the next step. It is the first time that a good truncation such as PML leads to a good DDM in the multi-way partition, which is essentially different from the popular recursive sweeping in the multi-way partition. Recently, we proposed and tested a novel diagonal sweeping domain decomposition method (DDM) with source transfer in in [59] with trace transfer in in [68]. The method improves the previous additive overlapping DDM by employing a more efficient subdomain solving order. We show that the method achieves the exact solution of the global PML problem with 2^n sweeps in the constant medium case. Although the sweeping usually implies sequential subdomain solves, the number of sequential steps required for each sweep in the method is only proportional to the n -th root of the number of subdomains when the domain decomposition is quasi-uniform with respect to all directions, thus it is very suitable for parallel computing of the Helmholtz problem with multiple right-hand sides through the pipeline processing.

Numerical solution of backward SDEs – Backward stochastic differential equation (BSDEs) and their solutions have been extensively studied by many researchers due to their applications especially in finance. Existence and uniqueness of solution of the BSDE were proved by Pardoux and Peng in 1990. We developed and analyzed a stable multi-step scheme on the time-space grids for numerically solving BSDEs in [16] and the decoupled forward-backward stochastic differential equations (FBSDEs) in [25]. In these schemes, the integrands, which are conditional mathematical expectations derived from the original equations, are approximated by the Lagrange interpolating polynomial with values of the integrands at multi-time levels. In particular, they are then numerically evaluated using the Gauss-Hermite quadrature rule and linear interpolation in the spatial grids.

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